QSAR-Guided Generative Design for TB Drug Discovery

Model Evaluation, Series C/E Results, and Generative Campaign Outputs

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TB QSAR Project

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Outline

1. QSAR Model Development
2. Series C Evaluation
3. Series E: Model Evaluation
4. ADME and Physicochemical Properties
5. Generative Modes
6. Gate 1 Shortlist
7. Source Mode Breakdown (200 compounds)
8. Summary

What We Built: A QSAR Predictive Model

Goal: Predict IC_{50} (potency vs. Mtb) from molecular structure alone

Input – Molecular Descriptors

- **RDKit** physicochemical descriptors (primary)
- **ECFP_r3** extended-connectivity fingerprints
- **MACCS** structural keys
- **Mordred** descriptor library

Output

$\hat{y} = \log_{10}(IC_{50}, \mu M)$
converted to pIC_{50} and IC_{50} for display

Workflow

1. Load Series C/E experimental data
2. Compute descriptor matrices
3. Train / cross-validate multiple regressors
4. Select best model (highest CV R^2)
5. Build ensemble for uncertainty
6. Save artifact `qsar_best_model.joblib`

Model Selection: Candidates and Results

Descriptor	Regressor	CV R^2	Role
RDKit	K-Nearest Neighbours (KNN)	**0.887**	Primary model
ECFP_r3	K-Nearest Neighbours (KNN)	0.677	Ensemble #1
RDKit	Decision Tree (DT)	0.665	Ensemble #2
RDKit	LightGBM (LGBM)	0.540	Ensemble #3
RDKit	Extra Trees (ET)	0.433	Ensemble #4

Why RDKit + KNN?

- Highest cross-validated R^2 on training data
- Robust to small dataset size (low-data QSAR regime)
- Interpretable: prediction is distance-weighted average of nearest neighbours

Ensemble Purpose

- Four additional models predict in parallel
- Uncertainty $= \sigma$ of ensemble predictions
- Flags structurally novel molecules where primary model may extrapolate

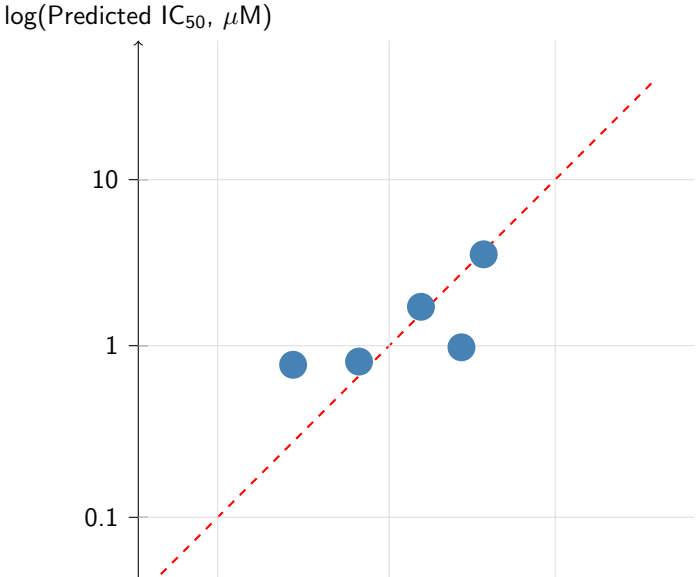
Series C: Test Data and Performance

Compound ID	Measured IC_{50} (μM)	Predicted IC_{50} (μM)	Error (fold)
ROY-0000173-001	4.697	4.984	1.06×
ROY-0000174-001	9.786	6.567	1.49×
ROY-0000220-001	0.781	1.173	1.50×
ROY-0000175-001	6.843	6.089	1.12×
ROY-0000211-001	1.800	1.233	1.46×
5 measured active compounds			

Performance Metrics

- **MAE** = 0.118 pIC_{50} units
- $\mathbf{R}^2 = 0.887$
- **Range**: 0.78 – 9.79 μM (measured)
- **Max fold error**: 1.50×

Series C: Measured vs. Predicted IC_{50} (log scale)



Series C: IC_{50} Comparison

Compound ID	Measured IC_{50} (μM)	Predicted IC_{50} (μM)	Series
ROY-0000173-001	4.697	4.98	C Spiro
ROY-0000174-001	9.786	6.57	C Spiro
ROY-0000220-001	0.781	1.17	C Spiro
ROY-0000221-001	>200	14.18	C Spiro
ROY-0000175-001	6.843	6.09	C Spiro
ROY-0000211-001	1.800	1.23	C Spiro

Key Observations

- Model correctly identifies ROY-0000220-001 and ROY-0000211-001 as most potent (sub-2 μM predicted)
- ROY-0000221-001 has measured IC_{50} >200 μM but predicted at 14.18 μM; structurally distinct (indole side-chain) — model underestimates inactivity for novel scaffolds
- Rank order of active compounds is well preserved

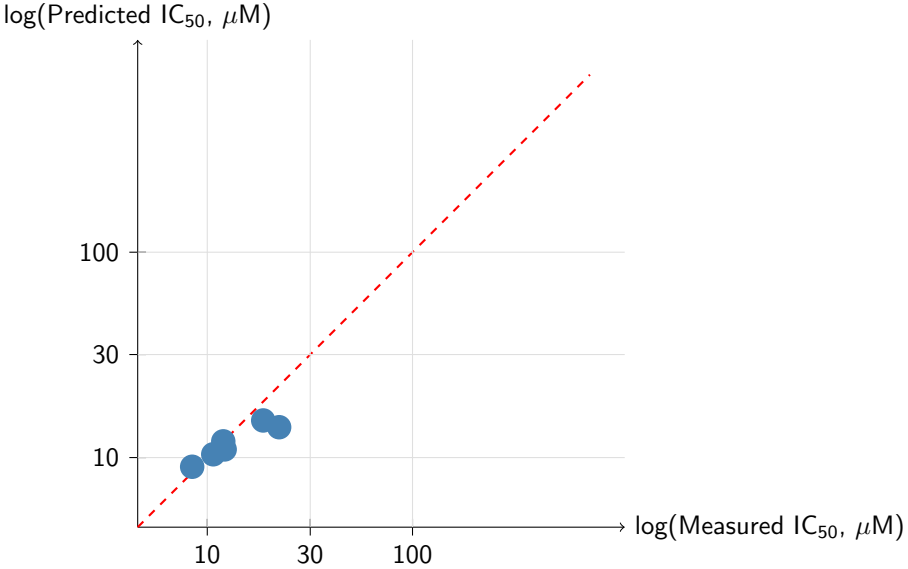
Series E: Test Data and Performance

Compound ID	Measured IC_{50} (μM)	Predicted IC_{50} (μM)	Error (fold)
ROY-0000308	17.80	13.08	1.36×
ROY-0000307	63.60	34.11	1.86×
ROY-0000287	6.07	7.38	1.22×
ROY-0000286	12.20	11.15	1.09×
ROY-0000275	108.00	27.33	3.95×
ROY-0000268	17.00	17.09	1.01×
6 measured test compounds			

Performance Metrics

- **Range**: 6.07 – 108.0 μM (measured)
- **Max error**: 3.95× (ROY-0000275, structural outlier)
- **Median error**: 1.23×
- **5 of 6** compounds: errors <1.9×

Series E: Measured vs. Predicted IC_{50} (log scale)



ADME Prediction Properties

Properties Predicted per Molecule

Property	Method
cLogP	RDKit Crippen
logS (solubility)	ESOL model
Microsomal CL	AZ QSAR model
Hepatocyte CL	AZ QSAR model
Half-life	Obach model
Human oral absorption	QikProp
Caco-2 permeability	QikProp

Gate 1 Shortlist ADME Summary

Property	Range
pred_cLogP	1.93 – 2.80
pred_logS_ESOL	−5.34 to −3.51
ADME indiv. score	0.70 – 0.94
passes_adme_indiv	65/200 (33%)
passes_metab. stab	65/200 (33%)

65 of 200 shortlisted compounds pass all ADME+metabolic stability filters — these are highest-priority compounds for synthesis

Generative Modes: Three Strategies

Mode	Description	Output Characteristic
Plain (Sampling)	Sample directly from the prior model without any optimization signal	Maximally diverse; explores full chemical space of the prior
Reinforcement Learning (RL)	Score each generated molecule; reward high-scoring structures to shift the distribution	Focused; converges on potency/ADME optima; risk of mode collapse
Curriculum Learning (CL)	Progressive reward shaping — start broad, tighten criteria over epochs	Balanced diversity and optimisation; avoids mode collapse

Molecular Transformation Modes

- **mol2mol_similarity** — generate analogues with Tanimoto constraint
- **mol2mol_scaffold** — preserve Murcko

Generator Families Used

reinvent	SMILES-based RNN
reinvent_pubchem	PubChem-pretrained
libinvent	R-group decoration

Generation Campaign: Mode × Family Matrix

Family	Plain	RL	CL	mol2mol
reinvent	✓	✓	✓	—
reinvent_pubchem	✓	✓	✓	similarity
libinvent	—	—	✓	—
libinvent_transformer_pubchem	—	—	✓	—
linkinvent	—	—	✓	—
linkinvent_transformer_pubchem	—	—	✓	similarity
pepinvent	✓	✓	—	—
nitro_bioisostere	✓	✓	✓	scaffold_generic

Master Pool Statistics

Total candidates	37,451
Unique scaffolds	14,287
Source families	8
Scoring models	QSAR + ADME

Scoring per Candidate

- pred_pIC50 (primary QSAR)
- pred_cLogP, pred_logS_ESOL
- Microsomal / hepatocyte clearance
- Tanimoto similarity to training set

Gate 1 Shortlist: 200 Top Candidates

Triage Score Formula

$$\text{triage} = w_1 \cdot \hat{p}\text{IC}_{50} + w_2 \cdot \text{ADME} + w_3 \cdot (1 - T_{\max})$$

Balances potency, drug-likeness, and novelty

Property	Min	Max
Triage score	0.770	0.921
pred pIC_{50}	4.94	6.11
pred IC_{50} (μM)	0.78	11.55
Tanimoto (to train)	0.28	0.86
pred cLogP	1.93	2.80
pred logS ESOL	−5.34	−3.51

Source Family (200 compounds)

Family	Count
reinvent_pubchem	137
nitro_bioisostere	53
reinvent	9
linkinvent_trans.	1

Source Mode (200 compounds)

Mode	Count
plain	140
mol2mol_similarity	28
mol2mol_scaffold_generic	22
curriculum_learning	6
mol2mol_scaffold	3

Gate 1 Shortlist: Unique Scaffolds

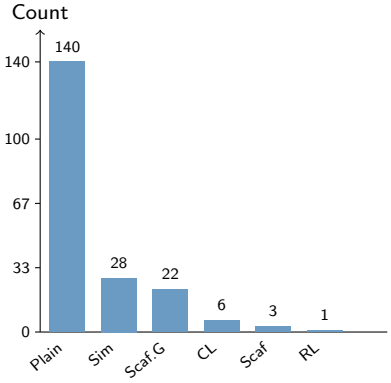
Scaffold Diversity

	Master Pool	Gate 1
Candidates	37,451	200
Unique scaffolds	14,287	155
Scaffold diversity	38.1%	77.5%

Gate 1 shortlist has **much higher scaffold diversity** than the master pool — the triage score effectively up-weights novel chemotypes.

Next Step

Compounds passing `passes_adme_individual` (65 of 200) are highest-priority for experimental synthesis and testing.



Source mode distribution of Gate 1 shortlisted compounds

Source Mode Breakdown: Data Source

- **Dataset:** `outputs/hybrid_stage/gate1/gate1_shortlist.csv`
- **Total compounds:** 200 (Gate 1 shortlist)
- **Sorting:** `pred_ic50_uM` ascending (low $\rightarrow$ high)
- **Columns shown:** source mode, source family, predicted IC_{50}, pIC_{50}, cLogP, SMILES
- **Rendered as:** 20 slides x 10 compounds per slide

Sorted export: `outputs/generated/_campaign_overview/gate1_shortlist_sorted_ic50.csv`

Source Mode (200 compounds): Structures sorted by IC_{50} (1–10)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
1		plain	reinvent_pubchem	0.78	6.11	2.44
2		plain	reinvent_pubchem	0.78	6.11	2.74
3		plain	reinvent_pubchem	0.78	6.11	2.44
4		plain	reinvent_pubchem	0.85	6.07	2.44
5		plain	reinvent_pubchem	0.85	6.07	2.44
6		plain	reinvent_pubchem	0.85	6.07	2.44
7		plain	reinvent_pubchem	0.85	6.07	2.44
8		plain	reinvent_pubchem	0.85	6.07	2.44
9		plain	reinvent_pubchem	0.85	6.07	2.62
10		plain	reinvent_pubchem	0.85	6.07	2.24

Source Mode (200 compounds): Structures sorted by IC_{50} (11–20)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
11		plain	reinvent_pubchem	0.85	6.07	2.64
12		plain	reinvent_pubchem	0.85	6.07	2.66
13		plain	reinvent_pubchem	0.85	6.07	2.64
14		plain	reinvent_pubchem	0.85	6.07	2.66
15		plain	reinvent_pubchem	0.85	6.07	2.66
16		plain	reinvent_pubchem	0.85	6.07	2.80
17		plain	reinvent_pubchem	0.85	6.07	2.64
18		plain	reinvent_pubchem	0.85	6.07	2.62
19		plain	reinvent_pubchem	0.85	6.07	2.37
20		plain	reinvent_pubchem	0.85	6.07	2.34

Source Mode (200 compounds): Structures sorted by IC_{50} (21–30)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
21		plain	reinvent_pubchem	0.85	6.07	2.37
22		plain	reinvent_pubchem	0.85	6.07	2.34
23		plain	reinvent_pubchem	0.85	6.07	2.21
24		plain	reinvent_pubchem	0.85	6.07	2.25
25		plain	reinvent_pubchem	0.85	6.07	2.18
26		plain	reinvent_pubchem	0.85	6.07	2.37
27		plain	reinvent_pubchem	0.85	6.07	2.38
28		plain	reinvent_pubchem	0.85	6.07	2.38
29		plain	reinvent_pubchem	0.85	6.07	2.37
30		plain	reinvent_pubchem	0.85	6.07	2.38

Source Mode (200 compounds): Structures sorted by IC_{50} (31–40)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
31		plain	reinvent_pubchem	0.85	6.07	2.37
32		plain	reinvent_pubchem	0.85	6.07	2.38
33		plain	reinvent_pubchem	0.95	6.02	2.62
34		plain	reinvent_pubchem	0.95	6.02	2.64
35		plain	reinvent_pubchem	1.04	5.98	2.13
36		plain	reinvent_pubchem	1.13	5.95	2.28
37		plain	reinvent_pubchem	1.24	5.91	2.59
38		plain	reinvent_pubchem	1.27	5.89	2.45
39		plain	reinvent_pubchem	1.31	5.88	2.49
40		plain	reinvent_pubchem	1.31	5.88	2.66

Source Mode (200 compounds): Structures sorted by IC_{50} (41–50)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
41		plain	reinvent_pubchem	1.31	5.88	2.66
42		plain	reinvent_pubchem	1.50	5.83	2.63
43		plain	reinvent_pubchem	1.54	5.81	2.67
44		plain	reinvent_pubchem	1.54	5.81	2.31
45		plain	reinvent_pubchem	1.54	5.81	2.29
46		plain	reinvent_pubchem	1.94	5.71	2.44
47		plain	reinvent_pubchem	1.94	5.71	2.44
48		plain	reinvent_pubchem	1.94	5.71	2.64
49		plain	reinvent_pubchem	1.94	5.71	2.44
50		plain	reinvent_pubchem	1.94	5.71	2.37

Source Mode (200 compounds): Structures sorted by IC_{50} (51–60)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
51		plain	reinvent_pubchem	1.94	5.71	2.23
52		plain	reinvent_pubchem	1.94	5.71	2.36
53		plain	reinvent_pubchem	1.94	5.71	2.62
54		plain	reinvent_pubchem	2.49	5.60	2.06
55		plain	reinvent_pubchem	2.49	5.60	2.27
56		plain	reinvent_pubchem	2.71	5.57	2.19
57		plain	reinvent_pubchem	2.73	5.56	1.94
58		plain	reinvent_pubchem	2.77	5.56	2.44
59		plain	reinvent_pubchem	2.88	5.54	2.48
60		plain	reinvent_pubchem	2.88	5.54	2.44

Source Mode (200 compounds): Structures sorted by IC_{50} (61–70)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
61		plain	reinvent_pubchem	2.88	5.54	2.38
62		plain	reinvent_pubchem	2.88	5.54	2.53
63		plain	reinvent_pubchem	2.91	5.54	2.33
64		plain	reinvent_pubchem	2.91	5.54	2.44
65		plain	reinvent_pubchem	3.14	5.50	1.93
66		plain	reinvent_pubchem	3.53	5.45	2.31
67		mol2mol_scaffold	nitro_bioisostere_campaign	3.61	5.44	2.25
68		plain	reinvent_pubchem	3.61	5.44	2.25
69		mol2mol_scaffold	nitro_bioisostere_campaign	3.61	5.44	2.04
70		plain	reinvent_pubchem	3.61	5.44	2.22

Source Mode (200 compounds): Structures sorted by IC_{50} (71–80)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
71		mol2mol_scaffold_generic	nitro_bioisostere_campaign	3.61	5.44	2.29
72		plain	reinvent_pubchem	3.76	5.42	2.30
73		plain	reinvent_pubchem	3.80	5.42	2.30
74		plain	reinvent_pubchem	3.84	5.42	2.39
75		plain	reinvent	3.96	5.40	2.42
76		mol2mol_similarity	nitro_bioisostere_campaign	4.03	5.39	2.33
77		plain	reinvent_pubchem	4.03	5.39	2.11
78		plain	reinvent_pubchem	4.03	5.39	2.11
79		plain	reinvent_pubchem	4.03	5.39	2.15
80		plain	reinvent_pubchem	4.03	5.39	2.18

Source Mode (200 compounds): Structures sorted by IC_{50} (81–90)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
81		plain	reinvent_pubchem	4.03	5.39	2.31
82		plain	reinvent_pubchem	4.03	5.39	2.21
83		plain	reinvent_pubchem	4.03	5.39	2.12
84		plain	reinvent_pubchem	4.03	5.39	2.45
85		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.03	5.39	2.35
86		plain	reinvent_pubchem	4.03	5.39	2.13
87		mol2mol_similarity	nitro_bioisostere_campaign	4.03	5.39	2.38
88		plain	reinvent_pubchem	4.03	5.39	2.04
89		mol2mol_similarity	nitro_bioisostere_campaign	4.03	5.39	2.06
90		plain	reinvent_pubchem	4.18	5.38	2.04

Source Mode (200 compounds): Structures sorted by IC_{50} (91–100)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
91		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.18	5.38	2.24
92		mol2mol_similarity	nitro_bioisostere_campaign	4.23	5.37	2.02
93		mol2mol_similarity	nitro_bioisostere_campaign	4.23	5.37	2.24
94		mol2mol_similarity	nitro_bioisostere_campaign	4.23	5.37	1.99
95		mol2mol_scaffold	nitro_bioisostere_campaign	4.24	5.37	2.30
96		mol2mol_similarity	nitro_bioisostere_campaign	4.24	5.37	2.41
97		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.24	5.37	2.17
98		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.24	5.37	2.16
99		curriculum_learning	reinvent	4.33	5.36	2.31
100		mol2mol_similarity	nitro_bioisostere_campaign	4.41	5.36	2.38

Source Mode (200 compounds): Structures sorted by IC_{50} (101–110)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
101		mol2mol_similarity	nitro_bioisostere_campaign	4.41	5.36	2.08
102		mol2mol_similarity	nitro_bioisostere_campaign	4.41	5.36	2.17
103		mol2mol_similarity	nitro_bioisostere_campaign	4.41	5.36	2.22
104		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.41	5.36	2.19
105		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.41	5.36	2.21
106		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.41	5.36	2.24
107		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.41	5.36	2.14
108		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.41	5.36	2.27
109		plain	reinvent_pubchem	4.44	5.35	2.05
110		plain	reinvent_pubchem	4.55	5.34	2.31

Source Mode (200 compounds): Structures sorted by IC_{50} (111–120)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
111		plain	reinvent_pubchem	4.55	5.34	2.23
112		mol2mol_similarity	nitro_bioisostere_campaign	4.58	5.34	2.34
113		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.64	5.33	2.25
114		mol2mol_similarity	nitro_bioisostere_campaign	4.70	5.33	2.27
115		mol2mol_similarity	nitro_bioisostere_campaign	4.71	5.33	2.28
116		mol2mol_similarity	nitro_bioisostere_campaign	4.72	5.33	2.04
117		curriculum_learning	reinvent	4.77	5.32	2.06
118		mol2mol_similarity	nitro_bioisostere_campaign	4.77	5.32	2.32
119		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.98	5.30	2.10
120		plain	reinvent_pubchem	4.98	5.30	2.27

Source Mode (200 compounds): Structures sorted by IC_{50} (121–130)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
121		mol2mol_scaffold_generic	nitro_bioisostere_campaign	4.98	5.30	2.10
122		mol2mol_similarity	nitro_bioisostere_campaign	4.98	5.30	2.33
123		plain	reinvent_pubchem	5.14	5.29	2.28
124		mol2mol_similarity	nitro_bioisostere_campaign	5.14	5.29	2.31
125		mol2mol_similarity	nitro_bioisostere_campaign	5.18	5.29	2.10
126		mol2mol_scaffold_generic	nitro_bioisostere_campaign	5.18	5.29	2.33
127		mol2mol_similarity	nitro_bioisostere_campaign	5.18	5.29	2.18
128		mol2mol_similarity	nitro_bioisostere_campaign	5.25	5.28	2.19
129		mol2mol_similarity	nitro_bioisostere_campaign	5.28	5.28	2.31
130		plain	reinvent_pubchem	5.31	5.28	2.31

Source Mode (200 compounds): Structures sorted by IC_{50} (131–140)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
131		plain	reinvent_pubchem	5.53	5.26	2.31
132		plain	reinvent_pubchem	5.53	5.26	2.31
133		plain	reinvent_pubchem	5.53	5.26	2.33
134		plain	reinvent_pubchem	5.53	5.26	2.33
135		plain	reinvent_pubchem	5.53	5.26	2.19
136		plain	reinvent_pubchem	5.53	5.26	2.19
137		plain	reinvent	5.55	5.26	2.22
138		mol2mol_similarity	nitro_bioisostere_campaign	5.57	5.25	2.15
139		mol2mol_similarity	nitro_bioisostere_campaign	5.57	5.25	2.19
140		curriculum_learning	reinvent	5.76	5.24	2.13

Source Mode (200 compounds): Structures sorted by IC_{50} (141–150)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
141		mol2mol_scaffold_generic	nitro_bioisostere_campaign	5.83	5.23	2.16
142		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.03	5.22	2.20
143		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.03	5.22	2.20
144		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.09	5.22	2.11
145		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.09	5.22	2.25
146		plain	reinvent_pubchem	6.21	5.21	2.26
147		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.28	5.20	2.26
148		mol2mol_similarity	nitro_bioisostere_campaign	6.28	5.20	2.19
149		mol2mol_similarity	nitro_bioisostere_campaign	6.28	5.20	2.25
150		plain	reinvent_pubchem	6.35	5.20	2.29

Source Mode (200 compounds): Structures sorted by IC_{50} (151–160)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
151		plain	reinvent	6.35	5.20	2.18
152		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.64	5.18	2.32
153		mol2mol_scaffold_generic	nitro_bioisostere_campaign	6.68	5.18	2.29
154		plain	reinvent_pubchem	7.38	5.13	2.31
155		mol2mol_similarity	nitro_bioisostere_campaign	7.38	5.13	2.22
156		plain	reinvent_pubchem	7.38	5.13	2.22
157		mol2mol_similarity	nitro_bioisostere_campaign	7.45	5.13	2.29
158		curriculum_learning	linkinvent_transformer_pubchem	7.52	5.12	2.21
159		plain	reinvent_pubchem	7.61	5.12	2.27
160		plain	reinvent_pubchem	7.61	5.12	2.31

Source Mode (200 compounds): Structures sorted by IC_{50} (161–170)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
161		plain	reinvent_pubchem	7.61	5.12	2.31
162		plain	reinvent_pubchem	7.61	5.12	2.27
163		plain	reinvent_pubchem	8.05	5.09	2.16
164		plain	reinvent_pubchem	8.17	5.09	2.20
165		plain	reinvent_pubchem	8.17	5.09	2.20
166		plain	reinvent_pubchem	8.17	5.09	2.13
167		plain	reinvent_pubchem	8.17	5.09	2.20
168		plain	reinvent_pubchem	8.17	5.09	2.20
169		plain	reinvent_pubchem	8.17	5.09	2.38
170		plain	reinvent_pubchem	8.17	5.09	2.31

Source Mode (200 compounds): Structures sorted by IC_{50} (171–180)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
171		plain	reinvent_pubchem	8.17	5.09	2.34
172		plain	reinvent_pubchem	8.17	5.09	2.34
173		plain	reinvent_pubchem	8.17	5.09	2.12
174		plain	reinvent_pubchem	8.17	5.09	2.20
175		plain	reinvent_pubchem	8.17	5.09	2.12
176		plain	reinvent_pubchem	8.36	5.08	2.28
177		reinforcement_learning	reinvent	8.97	5.05	2.20
178		curriculum_learning	reinvent	9.12	5.04	2.17
179		plain	reinvent_pubchem	9.36	5.03	2.16
180		plain	reinvent_pubchem	9.40	5.03	2.28

Source Mode (200 compounds): Structures sorted by IC_{50} (181–190)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
181		plain	reinvent_pubchem	9.80	5.01	2.18
182		plain	reinvent_pubchem	9.80	5.01	2.18
183		plain	reinvent_pubchem	9.81	5.01	2.28
184		curriculum_learning	reinvent	9.81	5.01	2.24
185		plain	reinvent_pubchem	10.14	4.99	2.23
186		plain	reinvent_pubchem	10.19	4.99	2.24
187		plain	reinvent_pubchem	10.35	4.98	2.26
188		plain	reinvent_pubchem	10.35	4.98	2.26
189		plain	reinvent_pubchem	10.48	4.98	2.17
190		plain	reinvent_pubchem	10.48	4.98	2.31

Source Mode (200 compounds): Structures sorted by IC_{50} (191–200)

Rank	Structure	Source mode	Family	IC_{50} (μM)	pIC_{50}	cLogP
191		plain	reinvent_pubchem	10.48	4.98	2.26
192		plain	reinvent_pubchem	10.48	4.98	2.20
193		plain	reinvent_pubchem	10.48	4.98	2.20
194		plain	reinvent_pubchem	10.48	4.98	2.18
195		plain	reinvent_pubchem	10.48	4.98	2.21
196		plain	reinvent_pubchem	10.48	4.98	2.24
197		plain	reinvent_pubchem	10.48	4.98	2.24
198		plain	reinvent_pubchem	10.48	4.98	2.26
199		plain	reinvent_pubchem	10.72	4.97	2.20
200		plain	reinvent_pubchem	11.55	4.94	2.28

Summary: QSAR Model Performance

Metric	Value	Context
Primary model	RDKit + KNN	Highest CV R^2
CV R^2	0.887	Excellent for low-data QSAR
Test MAE (Series C)	0.118 pIC_{50}	≈32% IC_{50} error
Test R^2 (Series C)	0.887	5 measured compounds
Series E predictions	7.38 – 34.11 μM	6 prospective compounds
Ensemble models	4 additional	Uncertainty estimation

Generation Campaign

Total candidates	37,451
Gate 1 shortlisted	200
Unique scaffolds (Gate 1)	155
ADME-passing	65 (33%)
Best pred IC_{50}	0.78 μM
Best triage score	0.921

Recommended Next Steps

1. Synthesise top 65 ADME-passing Gate 1 compounds
2. Measure IC_{50} vs. Mtb H37Rv
3. Re-train QSAR with new data (Series C + E + Gate 1 actives)
4. Iterate: run new generative campaign

All Prediction Measurements at a Glance

Measurement	Gate 1 Range	Description
pred_pIC50	4.94 – 6.11	Primary potency prediction
pred_ic50_uM	0.78 – 11.55 μM	IC_{50} in micromolar
pred_cLogP	1.93 – 2.80	Lipophilicity (Crippen)
pred_logS_ESOL	-5.34 to -3.51	Aqueous solubility (log mol/L)
pred_MetStab_CL_Microsome	—	Microsomal clearance (AZ model)
pred_MetStab_CL_Hepatocyte	—	Hepatocyte clearance (AZ model)
adme_individual_score	0.70 – 0.94	Composite ADME scoring
triage_score	0.77 – 0.92	Final composite ranking
Tanimoto	0.28 – 0.86	Similarity to training set
pred_pIC50_uncertainty	0 (current run)	Ensemble std (σ predictions)

All predictions generated by the primary RDKit+KNN QSAR model and AZ ADME models. IC_{50} uncertainty requires a re-run with full ensemble enabled.

Thank You

QSAR-Guided Generative Design for TB Drug Discovery

March 2026

Primary model:	RDKit + KNN, $R^2 = 0.887$
Test MAE:	0.118 pIC_{50} (Series C)
Shortlisted:	200 compounds from 37,451
Best predicted:	0.78 μM IC_{50}